

3 (a) (i) $a = \frac{v - u}{t} = \frac{24.72 - 0}{4.5} = 5.493 \text{ m s}^{-2}$ [1]

$P_{\text{ave}} = F(\frac{1}{2}v_{\text{max}}) = ma(\frac{1}{2}v_{\text{max}}) = (6500)(5.493)(\frac{1}{2})(24.72) = 441 \times 10^3 \text{ W}$ [1]
(accept P = change in KE/ time)

(ii) Loss in KE = Gain in GPE
 $\frac{1}{2}mv^2 = mgh$
 $h = \frac{v^2}{2g} = \frac{24.72^2}{2(9.81)} = 31.2 \text{ m}$ [1]

Hence it will not go beyond the track which is 45.0 m tall.
 (accept comparing KE and GPE)

(iii) Let the speed of the train at the top of the loop be v_{top} ,
 $\frac{1}{2}mv_{\text{top}}^2 = \frac{1}{2}mv^2 - mgh$
 $v_{\text{top}} = 12.6 \text{ m s}^{-1}$ [1]

To stay on the seat (and not fall off) at the top of the inner loop, $N > 0$.

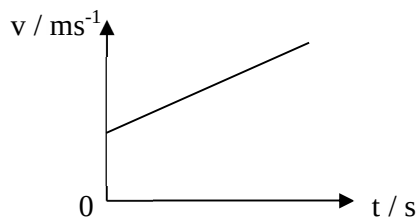
$N + mg = \frac{mv_{\text{min}}^2}{r}, \quad N = \frac{mv_{\text{min}}^2}{r} - mg$

Hence, to not fall off, $N > 0$ and therefore $v_{\text{min}} > \sqrt{rg}$.

$\sqrt{rg} = \sqrt{(11.5)(9.81)} = 10.6 \text{ m s}^{-1}$ [1]
 (accept calculating N instead)

Since $v_{\text{top}} > 10.6 \text{ m s}^{-1}$, the rider will not fall off if the lap-bar mechanism fail. [1]

(b) $v = r\omega \rightarrow v \propto r \propto t$ (since ω is constant and r is increasing at a steady rate)



Straight line [1]

Non-zero start and increasing [1]

(c) $m r \omega_{\text{max}}^2 = \frac{1}{2} mg$ [1]

$\rightarrow \omega_{\text{max}} = \sqrt{\frac{g}{2r}}$ [1]